

A coefficient normalization for positive definite forms

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Abstract

Every positive definite real homogeneous form admits linear coordinates in which its symmetric ordered coefficients have diagonal entries equal to one and all mixed entries strictly between zero and one. This answers Brandes’s question in Problem 11 of the 2019 Oberwolfach report on analytic number theory, including its absolute-value inequality. The proof chooses a basis close to a direction minimizing the form on a Euclidean sphere. The desired coefficient inequalities then follow from a positive quadratic term in a polynomial difference.

1 Statement and coefficient convention

Let $p: \mathbb{R}^m \rightarrow \mathbb{R}$ be a homogeneous polynomial of positive degree d , with $p(x) > 0$ for $x \neq 0$. Then d is even and $d \geq 2$. Let A be the unique symmetric d -linear form satisfying $p(x) = A(x, \dots, x)$. Throughout, a coefficient means a *symmetric ordered coefficient*:

$$p(t) = \sum_{j_1, \dots, j_d=1}^m A(e_{j_1}, \dots, e_{j_d}) t_{j_1} \cdots t_{j_d}.$$

Thus the ordinary coefficient of t^α is $\binom{d}{\alpha} A(e_1^{\alpha_1}, \dots, e_m^{\alpha_m})$. Repeated arguments are denoted by superscripts inside A .

Brandes asked whether a real invertible linear change of variables can make each coefficient’s absolute value at most the geometric mean of its corresponding diagonal entries [2, p. 3182, Problem 11]. The earlier discussion of pseudo-diagonal forms in [1] includes the quadratic case in Lemma 2.1. We prove the general basis-existence assertion with the ordered coefficient convention above.

Theorem 1. *Let $m \geq 1$ and let p be a positive definite real homogeneous form of degree $d \geq 2$. There is a basis $z_1, \dots, z_m$ of $\mathbb{R}^m$ such that*

$$p(z_i) = 1 \quad (1 \leq i \leq m), \quad 0 < A(z_{j_1}, \dots, z_{j_d}) < 1$$

whenever the indices $j_1, \dots, j_d$ are not all equal. In particular, in these coordinates the requested absolute-value bound holds for every tuple, with equality precisely for the pure tuples.

The change of variables is in $\mathrm{GL}_m(\mathbb{R})$; no orthogonality or quantitative conditioning constraint is imposed. For $m = 1$ the assertion about mixed tuples is vacuous.

2 Proof

Fix an auxiliary Euclidean norm. Choose a unit vector v where p attains its sphere minimum $c > 0$, and first replace p, A by $p/c, A/c$. The coefficient inequalities are invariant under this positive common scaling. Hence for the intermediate construction we have

$$p(v) = 1, \quad p(x) \geq \|x\|^d \quad (x \in \mathbb{R}^m). \quad (1)$$

Write $Q(a, b) = A(v^{d-2}, a, b)$.

Lemma 1. *For every $u \perp v$,*

$$A(v^{d-1}, u) = 0, \quad Q(u, u) \geq \frac{\|u\|^2}{d-1}.$$

Proof. By (1), the function $p(v + tu) - (1 + t^2\|u\|^2)^{d/2}$ has a minimum of zero at $t = 0$. Its first derivative is $dA(v^{d-1}, u)$, and its second derivative is $d(d-1)Q(u, u) - d\|u\|^2$. They are respectively zero and nonnegative, as claimed. $\square$

Choose an orthonormal basis $w_1, \dots, w_{m-1}$ of $v^\perp$, set $w_m = 0$, and define

$$x_i(\varepsilon) = v + \varepsilon w_i \quad (1 \leq i \leq m). \quad (2)$$

For every $\varepsilon \neq 0$ these vectors are independent: the tangential component of $\sum_i a_i x_i = 0$ forces $a_1 = \dots = a_{m-1} = 0$, and then the radial component forces $a_m = 0$.

For a fixed ordered tuple $J = (j_1, \dots, j_d)$ put $a_r = w_{j_r}$ and

$$N_J(\varepsilon) = A(v + \varepsilon a_1, \dots, v + \varepsilon a_d), \quad F_J(\varepsilon) = \prod_{r=1}^d p(v + \varepsilon a_r) - N_J(\varepsilon)^d.$$

Both expressions are polynomials in ε . Lemma 1 and multilinearity give

$$\begin{aligned} N_J(\varepsilon) &= 1 + \varepsilon^2 \sum_{r < s} Q(a_r, a_s) + O(\varepsilon^3), \\ p(v + \varepsilon a_r) &= 1 + \binom{d}{2} \varepsilon^2 Q(a_r, a_r) + O(\varepsilon^3). \end{aligned}$$

Consequently

$$\begin{aligned} F_J(\varepsilon) &= \varepsilon^2 \left(\frac{d(d-1)}{2} \sum_r Q(a_r, a_r) - d \sum_{r < s} Q(a_r, a_s) \right) + O(\varepsilon^3) \\ &= \frac{d}{2} \varepsilon^2 \sum_{r < s} Q(a_r - a_s, a_r - a_s) + O(\varepsilon^3). \end{aligned} \quad (3)$$

The last equality follows by expanding each pair difference; each diagonal term occurs in $d-1$ pairs.

If J is mixed, some $a_r - a_s$ is a nonzero vector in $v^\perp$. Lemma 1 makes the coefficient of ε^2 in (3) strictly positive. Thus $F_J(\varepsilon) > 0$ for all sufficiently small positive ε . There are only finitely many tuples, so one positive ε works for all mixed tuples. For a pure tuple, F_J is identically zero. Also $N_J(0) = 1$ for every tuple; by continuity and finiteness, decreasing ε if needed ensures $N_J(\varepsilon) > 0$ for all tuples.

Since d is even, these conclusions imply

$$0 < A(x_{j_1}, \dots, x_{j_d}) \leq \prod_{r=1}^d p(x_{j_r})^{1/d},$$

strictly for mixed tuples. Restore the original p and A : both sides scale by c , so the inequality is unchanged. Finally set $z_i = x_i/p(x_i)^{1/d}$ using the original p . Positive column rescaling preserves independence, makes $p(z_i) = 1$, and proves Theorem 1. The matrix S with columns z_i has $p'(t) = p(St)$ and polarization $A'(e_{j_1}, \dots, e_{j_d}) = A(z_{j_1}, \dots, z_{j_d})$, as required. $\square$

3 Scope and verification

The basis in (2) may be very close to singular. This is allowed by the question. Strict inequalities and nonzero determinant are open conditions, so the unnormalized basis x_i can also be chosen rational by density of $\mathbb{Q}^{m \times m}$ in $\mathbb{R}^{m \times m}$. The final unit-diagonal normalization need not preserve rationality. No claim is made here for merely nonnegative forms or for unimodular integral changes of variables.

The conclusion concerns the existence of suitable coordinates. For example, $p(x, y) = x^4 + 12x^2y^2 + y^4$ is positive definite but has $A(e_1, e_1, e_2, e_2) = 2 > 1$ in the original coordinates. The multinomial factor cannot be dropped when translating to ordinary monomial coefficients.

The accompanying open package contains an independent proof using positive definite bilinear slices and Cauchy–Schwarz, internal adversarial reviews, a dated priority audit, and a Python standard-library verifier. The verifier checks 12 rational examples exactly, including a nonconvex positive definite quartic, and cross-checks coefficients by polarization. These finite checks supplement the analytic proof; they are not a formal verification of the universal theorem. No separate computational assumption is needed in the proof.

Provenance and status. This is an unrefereed preprint. Generative AI supplied the initial candidate argument and assisted the internal independent checks, literature search, and writing. A bounded public-literature search completed on 23 September 2026 found no earlier general resolution; it does not establish first priority. The two direct antecedents are cited below. Paper, source, and verification materials are available at [the project page](#).

References

- [1] J. Brandes, *Forms representing forms: the definite case*, J. London Math. Soc. (2) **92** (2015), no. 2, 393–410. [doi:10.1112/jlms/jdv028](#).
- [2] H. L. Montgomery, *Problem Session*, in *Analytic Number Theory*, Oberwolfach Reports **16** (2019), no. 4, Report 50, pp. 3179–3183; Problem 11 (Brandes), p. 3182. [doi:10.4171/OWR/2019/50](#).